LAB 4: Interaction Design

Ms. Sana Ashraf

Interaction Design

Focuses on creating meaningful and engaging experiences for users. Design intuitive and effective interactions.

Types of Interaction Design Patterns

1) **Direct Manipulation:** involves allowing users to make directly interact with objects or elements on the screen, such as **dragging, tapping, or swiping**. Direct manipulation can enhance user's sense of control and provide immediate feedback.
2) **Conversational Interfaces:** This pattern focuses on designing interactions that mimic natural conversations between **users and digital systems**, such as **chatbots or voice assistants**. Learn the principles and considerations of designing conversational interfaces, including **language understanding, context & user flow**.
3) **Gamification:** Gamification is adding game mechanics into nongame context or environments (like website, online community, learning management system) to enhance user engagement and motivation and to increase participation. It create a sense of fun and enjoyment. Example are;
   Duolingo — streaks for daily lessons
   LinkedIn — progress bar for profile completion
   Starbucks — rewards for frequent purchases
4) **Creating Effective Interactions:** create effective interactions for websites or apps.

Some of the key concepts that will be covered include:

Affordances: This refers to the perceived actions that users can take based on the visual or interactive characteristics of element. Design interfaces that communicate affordance effectively, allowing users to understand the available actions or interactions.

Feedback: design appropriate feedback mechanisms, such as visual cues, animations, or notifications, to inform users about the system's response to their actions.

Constraints: show how to use constraints to guide user behavior and prevent errors or unintended actions. They will understand the role of physical, logical, and cultural constraints in designing intuitive and user-friendly interactions.

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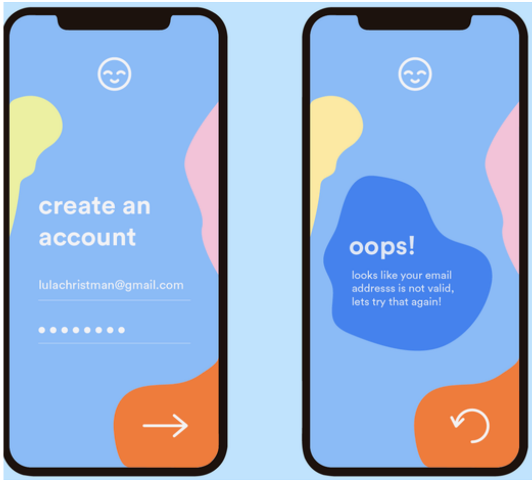

Tasks:

1) **Develop an interaction design for a website or app:**

   In this lab task, students will have the opportunity to apply their knowledge of interaction design principles and techniques to create an interactive design for a website or app. They will work on defining the user goals and tasks for the website or app and develop a design that provides intuitive and effective interactions to accomplish those goals. Students will consider factors such as user flow, navigation, input methods, feedback mechanisms, and visual design elements to create a cohesive and engaging user experience.

   To complete this task, students can use design tools and software to create wireframes, interactive prototypes, or mockups of the website or app.

2) **Test the effectiveness of the interaction design through usability testing:**

   Once the interaction design is developed, students will conduct usability testing to evaluate the effectiveness and usability of their design. Usability testing involves observing users as they interact with the design and collecting feedback and insights on their experience. Students will learn how to plan and conduct usability tests, define tasks for users to complete, and gather qualitative and quantitative data on user interactions and perceptions.

   During the usability testing, students will observe how users navigate the website or app, identify usability issues, and gather feedback on the design's strengths and weaknesses. This will help them understand how well their design meets user needs, identifies areas for improvement, and validates design decisions.

3) **Revise the interaction design based on feedback from users**

Based on the insights and feedback gathered from the usability testing, students will revise and refine their interaction design. They will analyze the usability test data, identify recurring issues or challenges faced by users, and consider the feedback received. Students will use this feedback to make informed design decisions and implement changes to improve the usability, effectiveness, and overall user experience of their design.

During the revision process, students may need to modify the user flow, navigation structure, visual design elements, or interaction patterns based on the identified issues. They will also consider how the feedback aligns with the principles and best practices of interaction design discussed in the lab. The goal is to iteratively improve the design and address any usability concerns raised by the users.